

16.1 Vector Fields

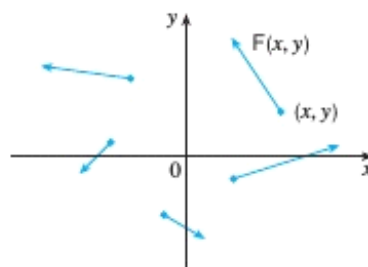
In general, a vector field is a function whose domain is a set of points in $\mathbb{R}^2$ (or $\mathbb{R}^3$) and whose range is a set of vectors in V_2 (or V_3).

Definition Let D be a set in $\mathbb{R}^2$ (a plane region). A vector field on $\mathbb{R}^2$ is a function F that assigns to each point (x, y) in D a two-dimensional vector $F(x, y)$.

The best way to picture a vector field is to draw the arrow representing the vector $F(x, y)$ starting at the point (x, y) . Since $F(x, y)$ is a two-dimensional vector, we can write it in terms of its component functions P and Q as follows:

$$F(x, y) = P(x, y) \mathbf{i} + Q(x, y) \mathbf{j} = \langle P(x, y), Q(x, y) \rangle$$

or, for short, $F = P \mathbf{i} + Q \mathbf{j}$



Definition Let E be a subset of $\mathbb{R}^3$. A vector field on $\mathbb{R}^3$ is a function F that assigns to each point (x, y, z) in E a three-dimensional vector $F(x, y, z)$.

Gradient Fields

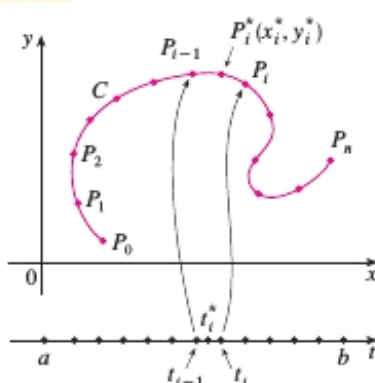
If f is a scalar function of two variables, recall from Section 14.6 that its gradient ∇f (or $\text{grad } f$) is defined by

$$\nabla f(x, y) = f_x(x, y) \mathbf{i} + f_y(x, y) \mathbf{j}$$

Therefore ∇f is really a vector field on $\mathbb{R}^2$ and is called a gradient vector field. Likewise, if f is a scalar function of three variables, its gradient is a vector field on $\mathbb{R}^3$ given by

$$\nabla f(x, y, z) = f_x(x, y, z) \mathbf{i} + f_y(x, y, z) \mathbf{j} + f_z(x, y, z) \mathbf{k}$$

16.2 Line Integrals



We start with a plane curve C given by the parametric equations

$$x = x(t) \quad y = y(t) \quad a \leq t \leq b$$

In Section 10.2 we found that the length of C is

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Definition If f is defined on a smooth curve C given by Equations 1, then the line integral of f along C is
$$\int_C f(x, y) ds = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*, y_i^*) \Delta s_i$$
 if this limit exists.

if f is a continuous function, then the limit always exists

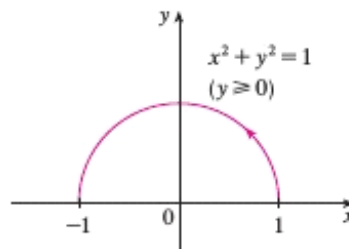
$$\int_C f(x, y) ds = \int_a^b f(x(t), y(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

EXAMPLE Evaluate $\int_C (2 + x^2 y) ds$, where C is the upper half of the unit circle $x^2 + y^2 = 1$.

SOLUTION we first need parametric equations to represent C . Recall that the unit circle can be parametrized by means of the equations $x = \cos t$ $y = \sin t$ and the upper half of the circle is described by the parameter interval $0 \leq t \leq \pi$. Therefore

$$\int_C (2 + x^2 y) ds = \int_0^\pi (2 + \cos^2 t \sin t) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$\dots \dots \dots = 2\pi + \frac{2}{3}$$



Two other line integrals are called the line integrals of f along C with respect to x and y :

$$\int_C f(x, y) dx = \int_a^b f(x(t), y(t)) x'(t) dt$$

$$\int_C f(x, y) dy = \int_a^b f(x(t), y(t)) y'(t) dt$$

Line Integrals in Space

We now suppose that C is a smooth space curve given by the parametric equations

$$x = x(t) \quad y = y(t) \quad z = z(t) \quad a \leq t \leq b$$

we define the line integral of f

along C (with respect to arc length) in a manner similar to that for plane curves:

$$\int_C f(x, y, z) ds = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*, y_i^*, z_i^*) \Delta s_i$$

$$\int_C f(x, y, z) ds = \int_a^b f(x(t), y(t), z(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$$

$$\int_a^b f(r(t)) |r'(t)| dt$$

For the special case $f(x, y, z) = 1$, we get

$$\int_C ds = \int_a^b |r'(t)| dt = L$$

where L is the length of the curve C

Line Integrals of Vector Fields

Recall from Section 6.4 that the work done by a variable force $f(x)$ in moving a particle from a to b along the x -axis is $W = \int_a^b f(x) dx$. Then in Section 12.3 we found that the work done by a constant force F in moving an object from a point P to another point Q in space is $W = F \cdot D$, where $D = \vec{PQ}$ is the displacement vector.

Now suppose that $F = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$ is a continuous force field on $\mathbb{R}^3$, such as the gravitational field of Example 4 in Section 16.1 or the electric force field of Example 5 in Section 16.1. (A force field on $\mathbb{R}^2$ could be regarded as a special case where $R = 0$ and P and Q depend only on x and y .) We wish to compute the work done by this force in moving a particle along a smooth curve C .

Therefore we define the work W done by the force field F as
$$W = \int_C F(x, y, z) \cdot T(x, y, z) ds = \int_C F \cdot T ds$$

If the curve C is given by the vector equation $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$, then $T(t) = \mathbf{r}'(t)/|\mathbf{r}'(t)|$, so

$$W = \int_a^b \left[F(\mathbf{r}(t)) \cdot \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} \right] |\mathbf{r}'(t)| dt = \int_a^b F(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$$

Definition Let F be a continuous vector field defined on a smooth curve C given by a vector function $\mathbf{r}(t)$, $a \leq t \leq b$. Then the line integral of F along C is

$$\int_C F \cdot d\mathbf{r} = \int_a^b F(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_C F \cdot T ds$$

EXAMPLE Find the work done by the force field $F(x, y) = x^2\mathbf{i} - xy\mathbf{j}$ in moving a particle along the quarter-circle $\mathbf{r}(t) = \cos t\mathbf{i} + \sin t\mathbf{j}$, $0 \leq t \leq \pi/2$.

SOLUTION Since $x = \cos t$ and $y = \sin t$, we have

$$F(\mathbf{r}(t)) = \cos^2 t\mathbf{i} - \cos t \sin t\mathbf{j}$$

$$\text{and } \mathbf{r}'(t) = -\sin t\mathbf{i} + \cos t\mathbf{j}$$

Therefore the work done is

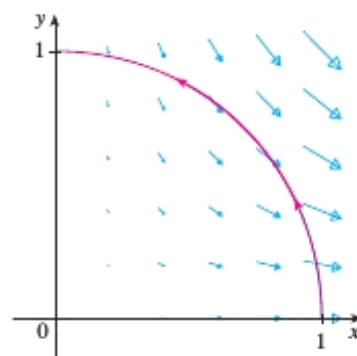
$$\begin{aligned} \int_C F \cdot d\mathbf{r} &= \int_0^{\pi/2} F(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt \\ &= \int_0^{\pi/2} (-2 \cos^2 t \sin t) dt \\ &= 2 \left[\frac{\cos^3 t}{3} \right]_0^{\pi/2} = -\frac{2}{3} \end{aligned}$$

NOTE Even though $\int_C F \cdot d\mathbf{r} = \int_C F \cdot T ds$ and integrals with respect to arc length are unchanged when orientation is reversed, it is still true that

$$\int_{-C} F \cdot d\mathbf{r} = -\int_C F \cdot d\mathbf{r}$$

because the unit tangent vector T is replaced by its negative when C is replaced by $-C$.

Figure shows the force field and the curve in Example. The work done is negative because the field impedes movement along the curve.



derivation notes:

the work done by the force $\mathbf{F}$ in moving the particle from P_{i-1} to

P_i is approximately $\mathbf{F}(x_i^*, y_i^*, z_i^*) \cdot [\Delta s_i \mathbf{T}(t_i^*)] = [\mathbf{F}(x_i^*, y_i^*, z_i^*) \cdot \mathbf{T}(t_i^*)] \Delta s_i$

and the total work done in moving the particle along C is approximately

$$\sum_{i=1}^n [\mathbf{F}(x_i^*, y_i^*, z_i^*) \cdot \mathbf{T}(x_i^*, y_i^*, z_i^*)] \Delta s_i = \int_C \mathbf{F} \cdot \mathbf{T} \, ds$$

Finally, we note the connection between line integrals of vector fields and line integrals of scalar fields. Suppose the vector field $\mathbf{F}$ on $\mathbb{R}^3$ is given in component form by the equation $\mathbf{F} = P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}$. We use Definition 13 to compute its line integral along C :

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) \, dt \\ &= \int_a^b (P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}) \cdot (x'(t) \mathbf{i} + y'(t) \mathbf{j} + z'(t) \mathbf{k}) \, dt \\ &= \int_a^b [P(x(t), y(t), z(t))x'(t) + Q(x(t), y(t), z(t))y'(t) + R(x(t), y(t), z(t))z'(t)] \, dt \end{aligned}$$

But this last integral is precisely the line integral

Therefore we have

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C P \, dx + Q \, dy + R \, dz \quad \text{where } \mathbf{F} = P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}$$

For example, the integral $\int_C y \, dx + z \, dy + x \, dz$ could be expressed as $\int_C \mathbf{F} \cdot d\mathbf{r}$ where

$$\mathbf{F}(x, y, z) = y \mathbf{i} + z \mathbf{j} + x \mathbf{k}$$

A vector field $\mathbf{F}$ is called a **conservative vector field** if it is the gradient of some scalar function, that is, if there exists a function f such that $\mathbf{F} = \nabla f$. In this situation f is called a **potential function** for $\mathbf{F}$.

Not all vector fields are conservative, but such fields do arise frequently in physics. For example, the gravitational field $\mathbf{F}$ is conservative because

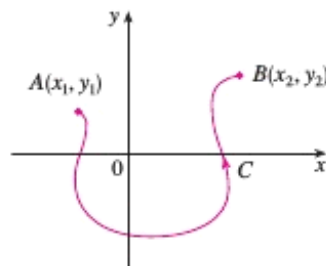
$$\begin{aligned} \text{if we define } f(x, y, z) &= \frac{mMG}{\sqrt{x^2 + y^2 + z^2}} \quad \text{then} \\ \mathbf{F}(x, y, z) &= \nabla f(x, y, z) = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k} \\ &= \frac{-mMGx}{(x^2 + y^2 + z^2)^{3/2}} \mathbf{i} + \frac{-mMGy}{(x^2 + y^2 + z^2)^{3/2}} \mathbf{j} + \frac{-mMGz}{(x^2 + y^2 + z^2)^{3/2}} \mathbf{k} \end{aligned}$$

16.3 The Fundamental Theorem for Line Integrals

Recall the Fundamental Theorem of Calculus can be written as

$$\int_a^b F'(x) dx = F(b) - F(a)$$

where F' is continuous on $[a, b]$.



If we think of the gradient vector ∇f of a function f of two or three variables as a sort of derivative of f , then the following theorem can be regarded as a version of the Fundamental Theorem for line integrals.

Theorem Let C be a smooth curve given by the vector function $\mathbf{r}(t)$, $a \leq t \leq b$. Let f be a differentiable function of two or three variables whose gradient vector ∇f is continuous on C . Then

$$\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$$

If f is a function of two variables and C is a plane curve with initial point $A(x_1, y_1)$ and terminal point $B(x_2, y_2)$, then

$$\int_C \nabla f \cdot d\mathbf{r} = f(x_2, y_2) - f(x_1, y_1)$$

If f is a function of three variables and C is a space curve joining the point $A(x_1, y_1, z_1)$ to the point $B(x_2, y_2, z_2)$, then we have

$$\int_C \nabla f \cdot d\mathbf{r} = f(x_2, y_2, z_2) - f(x_1, y_1, z_1)$$

Independence of Path

Suppose C_1 and C_2 are two piecewise-smooth curves (which are called paths) that have the same initial point A and terminal point B .

in general, $\int_{C_1} \mathbf{F} \cdot d\mathbf{r} \neq \int_{C_2} \mathbf{F} \cdot d\mathbf{r}$. But

$$\int_{C_1} \nabla f \cdot d\mathbf{r} = \int_{C_2} \nabla f \cdot d\mathbf{r}$$

whenever ∇f is continuous. In other words, the line integral of a conservative vector field depends only on the initial point and terminal point of a curve.

In general, if $\mathbf{F}$ is a continuous vector field with domain D , we say that the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path if $\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_{C_2} \mathbf{F} \cdot d\mathbf{r}$ for any two paths C_1 and C_2 in D that have the same initial and terminal points. With this terminology we can say that line integrals of conservative vector fields are independent of path.

Two Facts (Theorems) and Two Criteria (Theorems)

Theorem $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path in D if and only if $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed path C in D .

Theorem Suppose $\mathbf{F}$ is a vector field that is continuous on an open connected region D . If $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path in D , then $\mathbf{F}$ is a conservative vector field on D ; that is, there exists a function f such that $\nabla f = \mathbf{F}$.

Theorem If $\mathbf{F}(x, y) = P(x, y) \mathbf{i} + Q(x, y) \mathbf{j}$ is a conservative vector field, where P and Q have continuous first-order partial derivatives on a domain D , then throughout D we have

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

The converse of the Theorem is true only for a special type of region.

we need the concept of a **simple curve**, which is a curve that doesn't intersect itself anywhere between its endpoints.

A **simply-connected region** in the plane is a connected region D such that every simple closed curve in D encloses only points that are in D .



simply-connected region



regions that are not simply-connected

intuitively speaking, a simply-connected region contains no hole and can't consist of two separate pieces

Theorem Let $\mathbf{F} = P \mathbf{i} + Q \mathbf{j}$ be a vector field on an open simply-connected region D . Suppose that P and Q have continuous first-order derivatives and

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

throughout D

Then $\mathbf{F}$ is conservative.